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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This is a large multicentre randomised controlled trial in different regions in China.
- The associations between folic acid supplementation, homocysteine (Hcy) levels and pregnancy outcomes will be analysed.
- Subgroup analysis will detect based on Hcy levels and history of pregnancy loss and will provide valuable insights.
- A limitation is the inherent challenge in blinding participants to the folic acid dosage, which may introduce potential performance bias.
- Participants will be women with previous pregnancy loss. This may limit generalisability to the general population.

Introduction

Pregnancy loss refers to the spontaneous termination of a pregnancy before the fetus reaches 24 weeks of gestation.[1] Recurrent pregnancy loss (RPL) is specifically defined as the occurrence of two or more pregnancy losses, affecting approximately 5% of women of reproductive age.[2] Globally, an estimated 23â€‰million miscarriages occur annually, equating to 44 miscarriages per minute.[3] Research indicates that a single pregnancy loss is an independent risk factor for subsequent miscarriages.[3] Moreover, beyond the physical impact, nearly 40% of affected women suffer from posttraumatic stress reactions, including chronic anxiety, depression and social withdrawal.[4] This psychological burden can intensify family strain and complicate reproductive health decisions.[5] Consequently, improving subsequent live birth outcomes for women with a history of pregnancy loss remains a major focus in obstetrical research. Despite extensive investigation, about half of all pregnancy losses have no clear cause,[1] underscoring the need for further exploration. Currently, only a subset of causes can be identified, including parental chromosomal abnormalities, uterine anatomical anomalies, antiphospholipid syndrome and endocrine disorders like thyroid dysfunction. However, due to our limited understanding of the underlying pathophysiology, there are no standardised treatment protocols for pregnancy loss.

Folic acid plays a crucial role in human metabolism. It is essential for DNA synthesis, repair and methylation processes and is critical for cell division and growth. Its importance is particularly pronounced during pregnancy, especially in the first trimester. Gaskins *et al*[6] reported that compared with pregnant women with an average folate intake of 241â€‰Âµg/day, those with a higher folic acid intake had a lower stillbirth rate at 20 weeks of gestation. This suggests that high folate intake (averaging 1000â€‰Âµg/day) during weeks 8â€“19 of gestation, as part of prenatal care, may reduce miscarriage risk. Dey *et al*[7] found that lower plasma folate levels were associated with higher miscarriage rates, as well as higher preterm birth rates. Martin *et al* demonstrated that low serum folate levels were significantly associated with spontaneous abortion, whereas high folate status correlated with a non-significant reduction in risk.[8 9] Recent studies also indicated that 800â€‰Âµg folate supplementation during pregnancy effectively reduces homocysteine (Hcy) levels, improves pregnancy outcomes after frozen-thawed embryo transfer and significantly lowers miscarriage rates.[10] Collectively, these studies indicated that maintaining adequate folate levels may be protective against pregnancy loss, while folate deficiency is consistently linked to an increased risk. Additionally, neural tube development in the fetus begins in the initial weeks post-conception; folic acid deficiency during this stage significantly increases the risk of fetal neural tube defects and folic acid supplementation further prevents fetal congenital anomalies[11] .

However, a standardised consensus on the optimal folic acid dosage for patients with a history of pregnancy loss remains elusive, highlighting the need for further research. This issue is particularly relevant due to ongoing debates about the potential risks associated with high-dose folate supplementation (â‰¥800â€‰Âµg/day). Studies found that continuous high-dose folate intake from preconception through the second trimester may be linked to an increased risk of maternal-fetal complications, such as hypertensive disorders of pregnancy (HDP) and gestational diabetes mellitus (GDM).[12 13] Therefore, understanding the impact of supplementing with 800â€‰Âµg of folic acid during the first trimester on pregnancy outcomes in this patient population is crucial. Despite the significance of this issue, there is currently a lack of multi-centre, large-sample randomised controlled trials (RCTs) specifically designed to investigate this question, underscoring the need for rigorous scientific inquiry. In China, women planning pregnancy are advised to take 0.4â€‰mg (400â€‰Âµg) of folic acid daily from 3 months pre-conception until 12 weeks of gestation.[14] Accordingly, the control group in this study received this standard 400â€‰Âµg supplementation. This recommendation is consistent with international guidelines, such as those from the US Public Health Service, which advises 0.4â€“0.8â€‰mg daily for all women of reproductive potential.[15] However, the optimal dosage for women with a history of pregnancy loss remains unclear. Additionally, a large-scale study reported a history of spontaneous abortion in 9.06% of women who had ever been pregnant among the 299â€‰582â€‰women who reported having ever been pregnant in China.[16]Data from Lanzhou, a key site in our study located in the northern region, showed an NTD prevalence of 1.5% (486 cases among 320â€‰787 perinatal infants) between 2014 and 2019.[17] Our study is designed as a multi-centre investigation, with clinical sites located across several distinct regions of China. This strategic approach ensures that our cohort includes participants from geographic areas with diverse dietary patternsâ€”encompassing the wheat-based diets prevalent in the north, the rice-based diets common in the south, and various transitional dietary cultures in between. By including populations from these different backgrounds, we have intentionally built substantial dietary and lifestyle heterogeneity directly into the study design. Consequently, the findings generated from this multi-centre cohort are not specific to a single geographic or dietary profile. Instead, they are positioned to provide insights that are applicable to and representative of a broad spectrum of Chinese women experiencing pregnancy loss. We are confident that the outcomes of this study will hold significant insight for understanding and addressing pregnancy loss across China. This study aims to address this gap by investigating the association between folic acid status and maternal/fetal outcomes. Beyond risk factors like inadequate folate intake and elevated Hcy, genetic impairments in folate metabolism can induce a functional folate deficiency by disrupting one-carbon metabolism (FOCM)[18] . Notably, MTHFR variants (C677T/A1298C/A66G), which reduce enzymatic activity and are associated with an increased risk of miscarriage.[19 20] Based on this, our study aims to test the hypothesis that 800 Î¼g folic acid supplementation improves the live birth rate in women with prior pregnancy loss. In this study, Hcy reductions and MTHFR variants will both be investigated. Therefore, firstly, this study will quantify dose-response relationships between folic acid supplementation (400/800 Î¼g/day) and serum Hcy levels in women with pregnancy loss; secondly, evaluate synergistic effects of MTHFR genotypes (C677T/A1298C/A66G) and thresholds for optimal Hcy levels in PL on pregnancy outcomes; thirdly, to clarify the relationship between folic acid supplementation and pregnancy outcomes.

Methods and analysis

The participant recruitment for this multi-centre RCT commenced in April 2025 and is projected to be completed by April 2028. The study is currently in the participant recruitment.

Study design

This is a randomised, open-label, controlled trial that will be conducted at multiple geographically diverse centres across China, including the northwest, northeast, southwest and southeast. The study flow chart is shown in figure 1.

Trial flow chart. BMI, body mass index; FA, folic acid.

Participants

All participants in this study will be recruited in several hospitals who have consulted doctors for a pregnancy test with at least one previous pregnancy loss in China. All participating sites will receive ethics approval before the recruitment. Participants will be told details about the specific study protocol, and informed consent will be signed if they are willing to join this study. Baseline information will be obtained by their doctors through asking for their pregnancy loss-related history. Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Randomisation

Participants will be randomised into different groups according to a computer-generated randomisation procedure. Serial numbers will be allocated to one of the two arms in a 1:1 ratio. A fixed-size block randomisation of 4 was used to ensure the balance between groups.

Blinding

This study will be conducted in open-label, the allocation will remain open to participants and their doctors. The laboratory technician conducting the analysis of plasma Hcy was blinded to the allocation of groups.

Study visits

Participants will be queried regarding their demographic characteristics, including age, history of previous pregnancy loss, parity, lifestyle factors such as smoking and alcohol consumption, and chronic disease history. Data on diet, lifestyle and other potential confounding factors collected at baseline will be used to characterise the study population. Some physical examination will also be performed at baseline, such as body mass index (BMI), which is derived by dividing the weight in kilograms by the square of height. A questionnaire will also be conducted to know the dietary and lifestyle of the participants. At enrolment, MTHFR genotyping will be conducted for the C677T, A1298C and A66G variants. After 4â€‰weeks, a second visit will be asked to test the Hcy levels and an ultrasound examination will be conducted to confirm the pregnancy. Finally, at 12 gestation weeks, a final visit will be scheduled (Hcy levels will be measured at each visit). Biosampling will be conducted at enrolment, facilitating comprehensive analysis of underlying mechanisms and characterisation of prospective novel interventions to be explored in subsequent research. Compliance assessment will also involve quantifying the number of returned unused tablets. Other medical care will be conducted as usual based on relevant guidelines during pregnancy.

Supplement composition and supplementation schedule

The intervention group A will receive supplement tablets containing 800â€‰Âµg folic acid; intervention group B containing 400â€‰Âµg folic acid. Folic acid will be provided by Beijing Scrianen Pharmaceutical Co.ï¼ŒLtd.

Inclusion criterion

Patients with at least one previous pregnancy loss, coming to a reproductive clinic for a positive hCG test (serum) in 4â€“5 gestation weeks (<6 weeks); (1) 18â€“40 years, (2) BMI is between 18.5 and 28 kg/m^2 and (3) Patients who express willingness to participate in this study and provide informed consent.

Exclusion criterion

(1) Women with thyroid disorders [Thyroid-Stimulating Hormone(TSH)ï¼ž4 mIU/L], (2) Positive for any antibody of antiphospholipid (APL) [anticardiolipin or anti-Î²2-glycoprotein I (GPI)] and lupus anticoagulant, (3) Women with uterine cavity abnormality that might explain pregnancy loss, (4) Abnormal parental karyotype and (5) Patients who are taking folic acid antagonist drugs (such as methotrexate, ethamine).

Primary outcome measures

The primary outcome measures will focus on the live birth. Live birth is defined as the delivery of a live fetus after 24 completed weeks of gestational age.

Secondary outcomes of the study are

(1) Ongoing pregnancy at 24 gestation weeks (this will be determined via ultrasonographic examination performed at or after the 24-week gestation), (2) Early pregnancy loss (pregnancy loss that occurs before 10 weeks of pregnancy, including biochemical pregnancy), (3) The reduction of Hcy levels (plasma Hcy levels will be measured using standardised and validated methods at their central laboratory. The reduction will be calculated as the relative change from baseline to the second and the last measurement and (4) Fetal malformation rate (major fetal malformations will be systematically screened for and diagnosed via a standardised second-trimester structural anomaly scan (typically between 18â€“24 weeks of gestation) performed by certified sonographers. Suspected anomalies will be reviewed by a panel of experts, including maternal-fetal medicine specialists and clinical geneticists. Postnatal paediatric examinations will be used to confirm diagnoses for all live births. The classification of malformations will follow the International Classification of Diseases criteria.

(5) Incidence of pregnancy complications (eg, preterm delivery (defined as delivery between 24 and <37 weeks of gestation), GDM, HDP, premature rupture of membranes (PROM), abnormal placentation, gestational anaemia, placental abruption, postpartum haemorrhage (PPH)). All pregnancy complications will be rigorously defined and ascertained through detailed review of prenatal and delivery medical records using pre-specified criteria: preterm delivery (24 to <37 weeks). GDM: diagnosis will be based on the results of a standard 75â€‰g oral glucose tolerance

test performed between 24â€“28 weeks. HDP: this includes gestational hypertension, pre-eclampsia and eclampsia. PROM is a condition in which the amniotic sac breaks *before* the onset of labour. Gestational Anaemia: defined as a haemoglobin concentration of <110â€‰g/L at any time during pregnancy, as measured in venous blood samples. Abnormal placentation (including placental abruption and placenta previa). PPH: defined as an estimated blood loss of â‰¥500â€‰mL for vaginal delivery or â‰¥1000â€‰mL for caesarean section within 24 hours of delivery.

(6) Other pregnancy outcomes (mode of delivery, neonatal birth weight, gestational age at delivery, neonatal sex). Mode of delivery: categorised as spontaneous vaginal delivery, operative vaginal delivery or caesarean section. Neonatal birth weight: measured in grams immediately after delivery using a calibrated electronic scale. Gestational age at delivery: calculated in completed weeks based on the best obstetric estimate (LMP and first-trimester ultrasound). Neonatal sex: recorded as recorded at birth. Pregnancy outcomes will be confirmed and documented through a combination of medical record review and direct follow-up with participants post-delivery.

Definitions

RPL is defined as at least two or more spontaneous termination of a pregnancy prior to fetal viability, encompassing all losses occurring from conception until 24 weeks gestation according to ESHRE Guideline. Early pregnancy loss is defined as the loss of a pregnancy before 10 weeks of gestation.

Sample calculation

Based on the published studies,[21] the live birth rate in cases of at least one previous pregnancy loss is 53%. A minimum increase of 10% in the live birth rate is regarded as clinically relevant. To detect a 10% increase, with a power of 90% at a significance level of 0.05, 507 cases in each group are required to detect or refute a 10% difference. Taking into account a possible dropout rate and protocol violation rate of 10%, 558 cases will be recruited in each group (with a total of 1116 participants). The sample size calculation was performed with PASS 15 (NCSS, Kaysville, Utah, USA).

Withdrawal of individual participants

Participants have the right to withdraw from this study at any time. The decision to withdraw will neither affect their conventional clinical treatments nor their relationship with clinicians.

Withdrawal criteria are:

1. The participant wishes to discontinue participation in the study.
2. An overall uptake level of intervention of less than 60%.
3. The participant experiences an adverse reaction that the investigator deems to have a causal relationship with the intervention.
4. The participant is withdrawn at the discretion of the investigator due to medical reasons.

Consent will be obtained to access the medical records of participants who withdraw from the study, allowing for a comparison of key outcome measures between withdrawn and studied participants.

Statistical analysis

All the statistical analyses will be conducted according to the intention-to-treat principle and per-protocol. The data that deviated from a normal distribution will be subjected to analysis following appropriate transformations such as natural logarithm or reciprocal transformation. The differences in baseline characteristics and changes in Hcy levels between baseline and every time point will be calculated by one-way analysis of variance (ANOVA), Kruskalâ€“Wallis test or Pearsonâ€™s Ï‡^{2} test as appropriate for continuous and categorical variables, respectively. Additionally, we will use linear regression models to assess the efficacy of the supplements at lowering Hcy levels at midline and endline, adjusted for baseline Hcy, age, BMI, previous pregnancy losses and other important confounders. Stratified analysis will also be performed to explore whether the changes in plasma Hcy concentrations in different supplement groups are different in Hcy status at baseline. Logistic regression analysis will be used to assess the association between folic acid and pregnancy outcomes, while adjusting for potential confounders. Baseline hcy will be included as a covariate in the primary regression models to adjust for its potential confounding effect on the relationship between folic acid dose and live birth.

Subgroup analysis

Subgroup analysis will be conducted between the two arms based on the Hcy levels (<15â€‰Î¼mol/L and â‰¥15â€‰Î¼mol/L), previous pregnancy losses (<3 times and â‰¥3 times) and BMI (<25â€‰kg/m^{2} and â‰¥25â€‰kg/m^{2}).

Data monitoring

A data monitoring committee will not be established for this study due to its minimal risk profile, non-invasive procedures and routine primary care setting. No interim analyses or stopping guidelines are planned, as the intervention is low-risk. Trial oversight will instead be managed by the core research team through scheduled site check-ins, training reviews and data audits to ensure protocol adherence and data integrity.

Safety considerations, safety monitoring and AE reporting

The known side effects of folic acid will be disclosed to all participants during informed consent prior to randomisation. All adverse events (AEs)â€”whether observed by investigators, volunteered by participants or unrelated to treatmentâ€”will be recorded. AEs are defined as any undesirable experience occurring during the trial, irrespective of suspected causality. An SAE is any untoward medical occurrence that: results in death; is life-threatening; requires hospitalisation/prolongs existing hospitalisation; causes persistent/significant disability; is a congenital anomaly or poses new safety risks to participants (eg, unexpected reaction outcomes).

Ethics and dissemination

This study will be conducted in accordance with the principles of the Declaration of Helsinki. This study has been approved by the Institutional Review Board of Lanzhou University Second Hospital (2025A-347). Any important protocol modifications, including changes to eligibility criteria, outcomes or analysis methods, will be promptly communicated to the ethics committee, trial registry and relevant institutional stakeholders to ensure transparency and compliance. Written informed consent will be obtained from each participant before randomisation. The results of the trial will be presented at scientific meetings and reported in publications. Study data will be securely stored by designated researchers at each centre. Access requires approval from the centreâ€™s lead investigator. The trial complies with all relevant regulations, standard operating procedures and data protection requirements. Each participant receives a unique de-identification code. Data will be electronically collected from medical records in a deidentified format. All identifiable medical information remains confidential and is disclosed only as legally required. Quality control staff implement strict measures to protect participant confidentiality, including identity and results. The final trial datasets will be accessible to study investigators and Research Ethics Committees at participating sites. Data and samples collected will be submitted to the core research team.

Footnotes

References